

DA-Session-9-DPP

Assignment Questions

Basics of Sampling

Question 1. Suppose you want to study customer satisfaction in a mall. Explain how you would use systematic sampling to select participants.

Answer:

To study customer satisfaction in a mall using **systematic sampling**, I would first create a list of all customers entering or leaving the mall during the survey period. Then, I would determine the sampling interval by dividing the total number of customers by the desired sample size.

For example, if approximately 1,000 customers visit the mall and I want to survey 100 customers, the sampling interval would be:

$$1000 \div 100 = 10$$

Next, I would randomly select a starting point between 1 and 10, say 5. Then, I would select every 10th customer after that, such as the 5th, 15th, 25th, 35th, and so on, until 100 participants are chosen.

This method is simple, efficient, and helps ensure that the sample is spread evenly across the customer population, making it more representative of overall customer satisfaction.

Question 2. A school has 5 grades, each with different numbers of students. Describe how you would use stratified sampling to estimate the average exam score of all students.

Answer:

To estimate the average exam score of all students using **stratified sampling**, I would divide the students into five groups (strata) based on their grade levels. Since each grade has a different number of students, I would select a sample from each grade in proportion to its size.

For example, if Grade 1 contains 20% of the total students, then 20% of the sample would be selected from Grade 1. This process would be repeated for all five grades. After selecting the students randomly from each grade, I would collect their exam scores and calculate the average score for each grade.

Finally, I would combine the results from all grades, taking into account their proportions in the school population, to estimate the overall average exam score.

This method ensures that every grade is adequately represented, resulting in a more accurate and reliable estimate of the average exam score for the entire school.

Question 3. A school has 800 students: 400 in junior grades, 250 in senior grades, and 150 in advanced grades. You want to survey 100 students, ensuring proportional representation from each group. How many students should be sampled from each grade?

Answer:

To ensure **proportional representation**, use the formula:

$$\text{Sample Size for Group} = \left(\frac{\text{Number of Students in Group}}{\text{Total Students}} \right) \times \text{Total Sample Size}$$

Total Students = 800

Total Sample Size = 100

Junior Grades

$$\left(\frac{400}{800} \right) \times 100 = 50$$

Senior Grades

$$\left(\frac{250}{800} \right) \times 100 = 31.25 \approx 31$$

Advanced Grades

$$\left(\frac{150}{800} \right) \times 100 = 18.75 \approx 19$$

Final Sample Allocation

Grade Group	Number of Students	Sample Size
Junior Grades	400	50
Senior Grades	250	31
Advanced Grades	150	19
Total	800	100

Answer: To maintain proportional representation, the survey should include **50 junior-grade students, 31 senior-grade students, and 19 advanced-grade students.**

Question 4. A store wants to survey its customers. It has a list of 1,000 customer transactions and wants to sample every 20th transaction. If the starting point is transaction #7, list the first 10 transactions in the sample.

Answer:

This is an example of **systematic sampling**, where every 20th transaction is selected after a random starting point.

- Starting transaction = 7
- Sampling interval = 20

The sampled transactions are:

1. 7
2. 27
3. 47
4. 67
5. 87
6. 107
7. 127
8. 147
9. 167
10. 187

Final Answer

Sample Number Transaction Number

1	7
2	27
3	47
4	67
5	87
6	107
7	127
8	147
9	167
10	187

Therefore, the first 10 transactions in the sample are: 7, 27, 47, 67, 87, 107, 127, 147, 167, and 187.

Question 5. A city has 200 neighbourhoods. You want to estimate the average electricity consumption per household. You select 10 neighbourhoods at random and survey all households in these neighbourhoods. What is the main advantage of cluster sampling in this scenario?

Answer: The primary advantage of cluster sampling is that it reduces the cost and time of data collection by concentrating the survey on a few randomly selected neighbourhoods rather than the entire city.

Standard Error Calculation

Question 6. In a population of students, the average study time is 10 hours per week, with a standard deviation of 3 hours. You take a sample of size 36. Calculate the standard error of the mean

Answer:

The **Standard Error of the Mean (SE)** is calculated using the formula:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where:

- $\sigma = 3$ hours (standard deviation)
- $n = 36$ (sample size)

Substituting the values:

$$\begin{aligned} SE &= \frac{3}{\sqrt{36}} \\ SE &= \frac{3}{6} \\ SE &= 0.5 \end{aligned}$$

Final Answer:

The standard error of the mean is 0.5 hours per week.

Question 7. If the factory's light bulb lifespans have a mean of 1,200 hours and a standard deviation of 200 hours, calculate the standard error for samples of size 50

Answer:

The **Standard Error (SE)** is calculated using the formula:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Where:

- $\sigma = 200$ hours (standard deviation)

- $n = 50$ (sample size)

Substituting the values:

$$SE = \frac{200}{\sqrt{50}}$$

$$SE = \frac{200}{7.071}$$

$$SE \approx 28.28$$

Final Answer:

The standard error for samples of size 50 is approximately 28.28 hours.

Effect of Sample Size on Variability

Question 8. A population has a mean of 50 and a standard deviation of 10. Compare the standard error for sample sizes of 16, 64, and 256. Discuss how increasing the sample size affects variability.

Answer:

The **Standard Error (SE)** is calculated using the formula:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Given:

- Population Mean (μ) = 50
- Standard Deviation (σ) = 10

For Sample Size $n = 16$

$$SE = \frac{10}{\sqrt{16}} = \frac{10}{4} = 2.5$$

For Sample Size $n = 64$

$$SE = \frac{10}{\sqrt{64}} = \frac{10}{8} = 1.25$$

For Sample Size $n = 256$

$$SE = \frac{10}{\sqrt{256}} = \frac{10}{16} = 0.625$$

Comparison Table

Sample Size (n) Standard Error (SE)

16	2.5
64	1.25
256	0.625

Discussion

As the sample size increases, the standard error decreases. A smaller standard error means less variability in the sample mean and a more precise estimate of the population mean. Therefore, larger sample sizes produce more reliable results and reduce sampling variability.

Final Answer

The standard errors are **2.5, 1.25, and 0.625** for sample sizes **16, 64, and 256**, respectively. Increasing the sample size decreases variability and improves the accuracy of the estimate.

Real-World Applications

Question 9. A company sends out a survey to 1,000 customers and receives responses from 150.

- Identify the type of sampling method used.
- Discuss potential biases in the sample.

Answer:

1. Identify the Type of Sampling Method Used

The sampling method used is **Voluntary Response Sampling (Self-Selected Sampling)** because customers choose whether or not to respond to the survey. Only those who are willing participate in the survey.

2. Potential Biases in the Sample

The sample may suffer from the following biases:

- **Non-Response Bias:** The 850 customers who did not respond may have different opinions from those who did respond.
- **Self-Selection Bias:** Customers with very positive or very negative experiences are more likely to respond, while neutral customers may ignore the survey.

- **Lack of Representativeness:** Since only 150 out of 1,000 customers responded, the sample may not accurately represent the views of all customers.

Question 10. A researcher collects 10 random samples of size 30 from a population. The means are 75, 78, 74, 76, 77, 75, 79, 76, 74, and 77. Create a sampling distribution of the mean and calculate its mean and standard deviation

Answer:

The **sampling distribution of the mean** is the distribution formed by the sample means collected from the population.

Given Sample Means

75, 78, 74, 76, 77, 75, 79, 76, 74, 77

Step 1: Create the Sampling Distribution

Sample Number Sample Mean

1	75
2	78
3	74
4	76
5	77
6	75
7	79
8	76
9	74
10	77

Step 2: Calculate the Mean of the Sampling Distribution

$$\begin{aligned}\bar{x} &= \frac{75 + 78 + 74 + 76 + 77 + 75 + 79 + 76 + 74 + 77}{10} \\ \bar{x} &= \frac{761}{10} \\ \bar{x} &= 76.1\end{aligned}$$

Step 3: Calculate the Standard Deviation

Using the sample means:

Mean Deviation from 76.1 Squared Deviation

75	-1.1	1.21
78	1.9	3.61

Mean Deviation from 76.1 Squared Deviation

74	-2.1	4.41
76	-0.1	0.01
77	0.9	0.81
75	-1.1	1.21
79	2.9	8.41
76	-0.1	0.01
74	-2.1	4.41
77	0.9	0.81

$$\sum(x - \bar{x})^2 = 24.90$$

$$SD = \sqrt{\frac{24.90}{10}}$$

$$SD = \sqrt{2.49}$$

$$SD \approx 1.58$$

Final Answer

- **Sampling Distribution:** 75, 78, 74, 76, 77, 75, 79, 76, 74, 77
- **Mean of Sampling Distribution = 76.1**
- **Standard Deviation ≈ 1.58**

Therefore, the sampling distribution has a **mean of 76.1** and a **standard deviation of approximately 1.58**.

Probability Using CLT

Question 11. The average weight of apples in an orchard is 200g with a standard deviation of 50g. A random sample of 36 apples is taken.

- What is the probability that the sample mean is greater than 210g

Answer:

Given:

- Population Mean (μ) = 200 g
- Population Standard Deviation (σ) = 50 g
- Sample Size (n) = 36

We need to find:

$$P(\bar{X} > 210)$$

Step 1: Calculate the Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE = \frac{50}{\sqrt{36}}$$

$$SE = \frac{50}{6}$$

$$SE = 8.33$$

Step 2: Calculate the Z-score

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{210 - 200}{8.33}$$

$$Z \approx 1.20$$

Step 3: Find the Probability

From the standard normal distribution table:

$$P(Z < 1.20) = 0.8849$$

Therefore,

$$P(Z > 1.20) = 1 - 0.8849$$

$$P(Z > 1.20) = 0.1151$$

Final Answer

$$P(\bar{X} > 210) = 0.1151$$

Therefore, the probability that the sample mean weight is greater than 210 g is approximately 0.1151, or 11.51%.

Thank you